# LibraryMember JavaBean & Security Answer Property

Assignment Problem 4

## Concept and approach

Use JavaBean naming exactly: getX/setX for ordinary properties and isPremiumMember/setPremiumMember for the boolean property. The membership ID setter is write-once. The security answer is write-only: retain only a deterministic one-way transformed value and do not provide any getter or other method that exposes it.

### Important implementation notes

membershipId changes only on the first valid setter call. A digest is a one-way representation; real password/security-answer storage should use a password-hashing scheme with a salt and work factor. The requirement here explicitly says no getter exists.

## Java solution

```
import java.nio.charset.StandardCharsets;
import java.security.MessageDigest;
import java.security.NoSuchAlgorithmException;

public class LibraryMember {
private String membershipId;
private String name;
private boolean premiumMember;
private byte[] securityAnswerDigest;

public LibraryMember() { }

public String getMembershipId() { return membershipId; }

public void setMembershipId(String id) {
if (membershipId == null && id != null && !id.isEmpty())
membershipId = id;
}

public String getName() { return name; }
public void setName(String name) { this.name = name; }

public boolean isPremiumMember() { return premiumMember; }
public void setPremiumMember(boolean premium) { this.premiumMember = premium; }

public void setSecurityAnswer(String answer) {
if (answer == null) {
securityAnswerDigest = null;
return;
}
try {
MessageDigest digest = MessageDigest.getInstance("SHA-256");
securityAnswerDigest = digest.digest(answer.getBytes(StandardCharsets.UTF_8));
} catch (NoSuchAlgorithmException e) {
throw new IllegalStateException("SHA-256 is unavailable", e);
}
}

// Deliberately no getter for securityAnswerDigest or securityAnswer.
}
```

## Sample output

```
LIB-8841
LIB-8841
true
(no observable output from setSecurityAnswer)
```